

InfoSec SIG @ KJSSE

FLAG OF THE DAY

FOTD #6 SOLUTION

Houston, We Have a Cat

CATEGORY	Forensics, Steganography
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DIFFICULTY	Medium-Hard
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DATE	22/09/2026
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FLAG	INFOSEC_SIG{M30W_I5_4L4G}
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LOOK PAST THE PICTURE

The attachment is a PNG of a cat drifting around a space station. Nothing looks off when you open it, and the filename is basically telling you the picture isn't the point. With PNG challenges, LSB steganography is one of the first things worth checking.

LSB stands for least significant bit. Data gets written into the last bit of each pixel's colour values, and flipping that bit changes a colour by one step out of 256. Your eyes can't catch that, so the image looks untouched.

Download the image from the Drive link, load it into the decoder at stylesuxx.github.io/steganography and hit Decode.

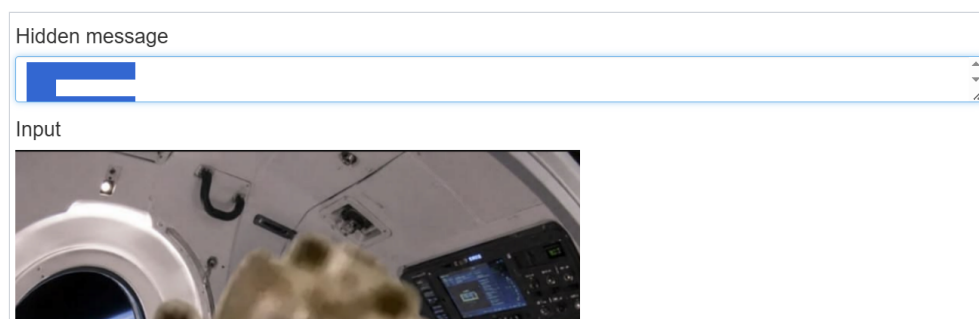


Figure 1. The decoder output looks blank until you select it.

Talk is cheap, Send flags.

The hidden message box seems empty, but selecting it shows there is content in there. It's made up entirely of spaces, tabs and line breaks.

READ THE SILENCE

A message built only from invisible characters points to Whitespace, an esoteric programming language where spaces, tabs and newlines are the only characters that matter. Everything else gets ignored.

Copy the output, open dCode's [Whitespace tool](#) and paste it into the ciphertext box. Remove the junk trailing after the real code, then click Decrypt.

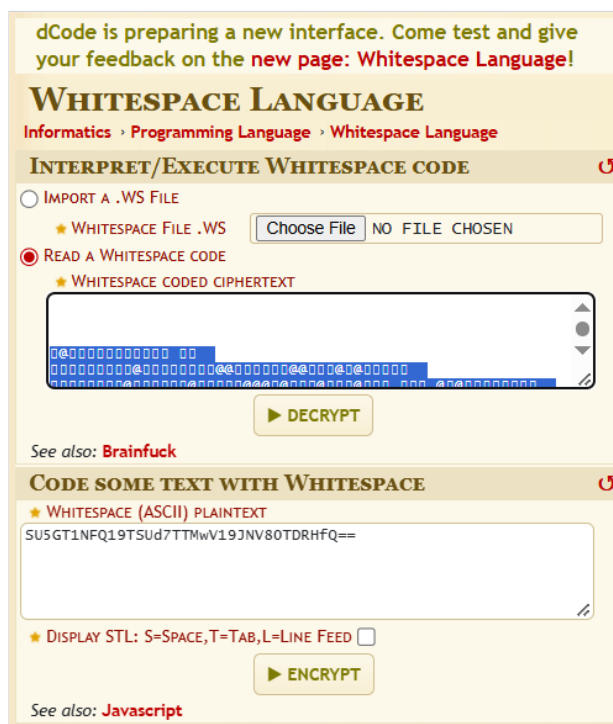


Figure 2. The whitespace code pasted into dCode.

The program prints this:

```
SU5GT1NFQ19TSUd7TTMwV19JNV80TDRHfQ==
```

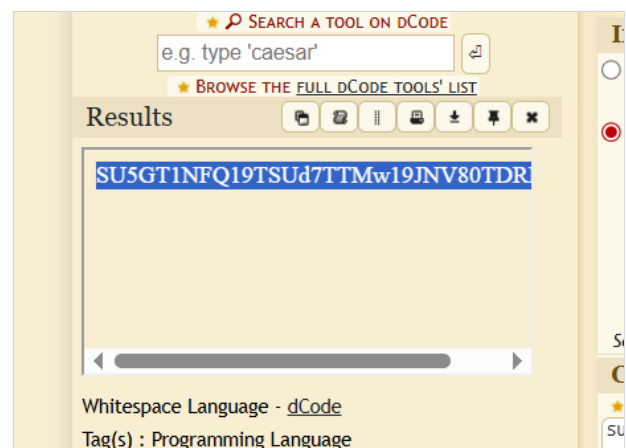


Figure 3. dCode runs the program and prints a string.

Talk is cheap, Send flags.

UNWRAP THE BASE64

The `==` at the end is Base64 padding, which gives it away. Decode it in a terminal:

```
$ echo "SU5GT1NFQ19TSUd7TTMwV19JNV80TDRHfQ==" | base64 -d  
INFOSEC_SIG{M30W_I5_4L4G}
```

Pasting it straight into Google works as well.

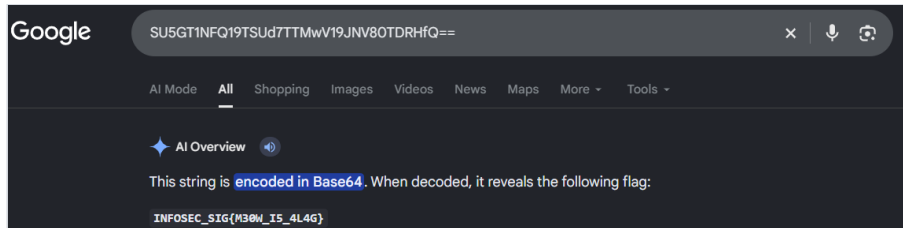


Figure 4. Google decoding the string.

INFOSEC_SIG{M30W_I5_4L4G}

TAKEAWAY

This one stacks three layers. LSB hides the data inside the pixels, Whitespace hides it in characters you can't see, and Base64 wraps the flag itself. If a tool hands you output that looks empty, select it or check its length before you decide it failed.

Talk is cheap, Send flags.